

NEURAL GRADIENTS ARE NEAR-LOGNORMAL: IMPROVED QUANTIZED AND SPARSE TRAINING

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ABSTRACT

While training can mostly be accelerated by reducing the time needed to propagate neural gradients back throughout the model, most previous works focus on the quantization/pruning of weights and activations. These methods are often not applicable to neural gradients, which have very different statistical properties. Distinguished from weights and activations, we find that the distribution of neural gradients is approximately lognormal. Considering this, we suggest two closed-form analytical methods to reduce the computational and memory burdens of neural gradients. The first method optimizes the floating-point format and scale of the gradients. The second method accurately sets sparsity thresholds for gradient pruning. Each method achieves state-of-the-art results on ImageNet. To the best of our knowledge, this paper is the first to (1) quantize the gradients to 6-bit floating-point formats, or (2) achieve up to 85% gradient sparsity — in each case without accuracy degradation. Reference implementation accompanies the paper.

1 INTRODUCTION

Neural gradients (gradients with respect to intermediate neural layer outputs) are used in the training process of deep networks to backpropagate the error-gradient throughout the model, thus allowing to compute the required weight updates. As these neural gradients are needed for a substantial ratio of the underlying computations (about $\frac{2}{3}$), compressing them can alleviate data-throughput requirements and accelerate the training process.

Many previous works (Banner et al., 2018b; Fang et al., 2020) compress tensors such as weights and activations by approximating their distributions using an analytically tractable density. These works often assume a bell-shaped distribution such as Gaussian or Laplace distributions, which have been reported to fail for neural gradients (Ye et al., 2019). One key observation in this paper is that neural gradient distributions are heavy-tailed, fundamentally different from the light-tailed distributions of weights and activations. Further statistical and distributional tests reveal gradient magnitudes follow a lognormal distribution.

Adopting this lognormal observation, our paper suggests two main applications — quantization and pruning, used to reduce the computational and memory burden of neural gradients. To tackle these challenges, we first formalize the problems and find closed-form expressions that enable us to predict the optimal quantization and pruning policies. These measures are easy to use and depend only on the estimated lognormal parameters.

In Figure 1 we summarize these applications and their derivation. The first application uses the lognormal prior to enabling low-precision floating-point (FP) quantization of the gradients. Here we optimize two tasks. The first task is to find a partition between mantissa and exponent bit-widths that minimizes quantization noise for a given n bit FP gradient representation. The second task is

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